# Health watch — sensor board

Sơ đồ phân cấp: mỗi ô là một trang con. Nhãn toàn cục (global label) nối tín hiệu giữa các trang; ký hiệu nguồn nối theo tên.

Ref 201–299. Sinh tự động bởi tools/gen_sch.py từ tools/design.py.

PPG AFE, LEDs, photodiodes, level shifter, skin temperature

File: health_watch_sensor_sensor.kicad_sch

Generated by tools/gen_sch.py from tools/design.py — edit the design file, not this sheet

**harisama1 — đồ án đồng hồ sức khỏe**

Sheet: /
File: health_watch_sensor.kicad_sch

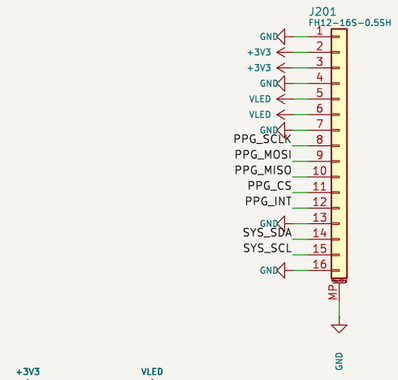
***Title: Health watch — sensor board — root***

| Size: A4 | Date: 2026-11-04 | **Rev: 1.0** |
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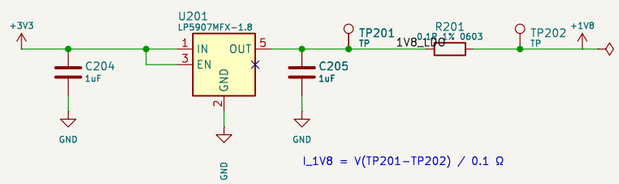
# PPG AFE, LEDs, photodiodes, level shifter, skin temperature

Bo cảm biến: LDO 1.8 V nhiều tiếp → MAX86141, LED SFH 7013 (xanh/đỏ/IR) và 2 photodiode nối thẳng vào AFE; SPI 1.8 V ↔ 3.3 V qua TXS0108E; MAX30208 nhiệt độ da.

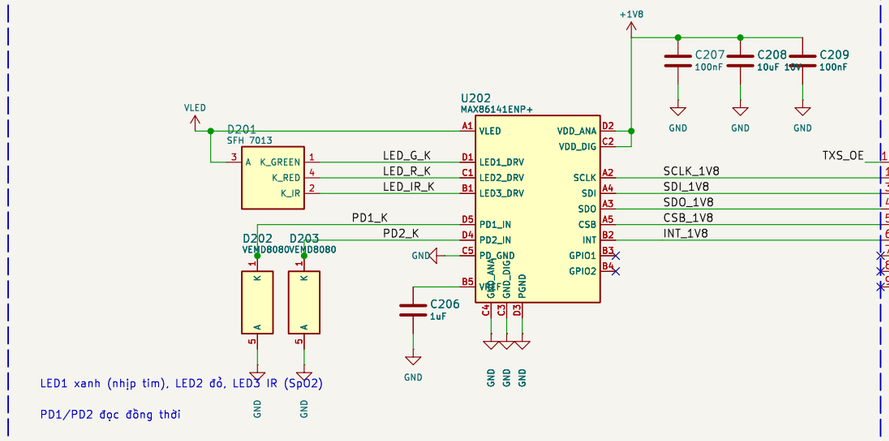
## FPC 16P nối bo chính



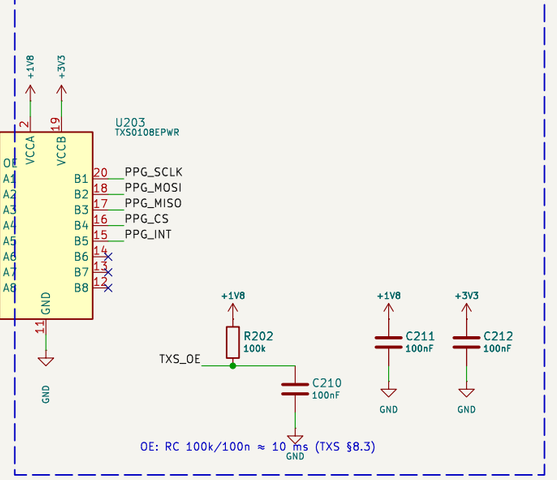

## LDO 1.8 V + shunt đo dòng



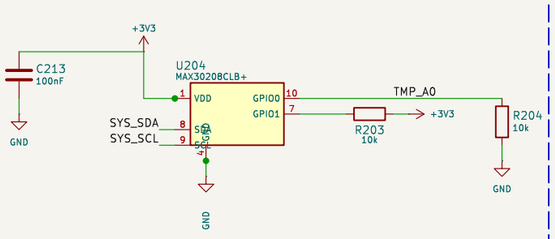

## MAX86141 + LED + photodiode



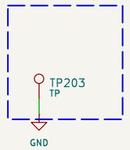

## Chuyển mức TXS0108E (1.8 ↔ 3.3 V)




## MAX30208 nhiệt độ da (I2C 0x52)




## Điểm đo / linh kiện phụ




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Sheet: /PPG AFE, LEDs, photodiodes, level shifter, skin temperature/

File: health_watch_sensor_sensor.kicad_sch

***Title: Health watch — sensor board — PPG AFE, LEDs, photodiodes, level shif***

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